

Exercise 4: Employee Management System

Arrays store elements in contiguous memory locations, which means all elements are placed one after another in memory. This allows direct access to any element using its index. The main advantage of arrays is fast access to elements because any position can be accessed in constant time. Arrays are also simple to implement and require less memory overhead compared to some other data structures.

2. Implementation

```
class Employee{
    int employeeID;
    String name;
    String position;
    double salary;
    public Employee(int employeeID, String name, String position, double
salary){
        this.employeeID = employeeID;
        this.name = name;
        this.position = position;
        this.salary = salary;
    }
}
class EmployeeManagementSystem{
    Employee[] employees;
    int count;
    public EmployeeManagementSystem(int size){
        employees = new Employee[size];
        count = 0;
    }
    public void addEmployee(Employee employee){
        if(count < employees.length){
            employees[count] = employee;
            count++;
        }
    }
    public Employee searchEmployee(int id){
        for(int i = 0; i < count; i++){
            if(employees[i].employeeID == id){
                return employees[i];
            }
        }
        return null;
    }
}
```

```

public void traverseEmployees(){
    for(int i = 0; i < count; i++){
        System.out.println(employees[i].employeeID + " " +
employees[i].name);
    }
}
public void deleteEmployee(int id){
    int index = -1;

    for(int i = 0; i < count; i++){
        if(employees[i].employeeID == id){
            index = i;
            break;
        }
    }
    if(index != -1){
        for(int i = index; i < count - 1; i++){
            employees[i] = employees[i + 1];
        }
        employees[count - 1] = null;
        count--;
    }
}
}

```

Analysis:

Add Employee:

- Best Case: $O(1)$
- Average Case: $O(1)$
- Worst Case: $O(1)$

Search Employee:

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Traverse Employees:

- Best Case: $O(n)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Delete Employee:

- Best Case: $O(1)$
- Average Case: $O(n)$
- Worst Case: $O(n)$

Limitations of Arrays:

Arrays have a fixed size, which means their capacity cannot be increased once they are created. Insertion and deletion operations can also be costly because elements may need to be shifted. Arrays are most suitable when the number of records is known in advance and frequent insertions or deletions are not required.